

Molecular Diagnostics
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Detecting disease at the molecular and genetic level

Molecular Diagnostics uses techniques that detect specific DNA, RNA, or protein markers to diagnose disease, identify infections, and guide personalized treatment decisions.

● DEFINITION

A collection of techniques used to analyze biological markers in the genome and proteome to diagnose and monitor disease.

● WHY IT MATTERS

Molecular diagnostics can detect disease earlier and more precisely than traditional methods, enabling faster treatment decisions and better patient outcomes.

● WORKING PRINCIPLE

INPUT / PROBLEM

CORE PROCESS

OUTCOME / APPLICATION

Molecular diagnostic tests detect specific genetic sequences or molecular markers associated with a disease or pathogen, often amplifying tiny amounts of genetic material to detectable levels for accurate diagnosis.

● QUICK FACTS

- ▶ PCR testing, central to molecular diagnostics, became a household term during the COVID-19 pandemic
- ▶ Liquid biopsies can sometimes detect cancer-related genetic material in blood before a tumor is visible on imaging

● KEY TERMS

- PCR
- Biomarker
- Next-generation sequencing
- Companion diagnostic

● CORE CONCEPTS

- ▶ PCR (polymerase chain reaction) testing
- ▶ Genetic biomarker detection
- ▶ Next-generation sequencing
- ▶ Companion diagnostics

● APPLICATIONS

- ▶ Infectious disease testing (PCR)
- ▶ Cancer genetic biomarker testing
- ▶ Inherited disease screening
- ▶ Personalized cancer treatment selection

● ADVANTAGES

- ▶ Highly sensitive and specific disease detection
- ▶ Enables personalized, targeted treatment decisions
- ▶ Can detect disease before symptoms appear

CAREER OPPORTUNITIES

Molecular diagnostics technologist, Clinical geneticist, Diagnostics R&D scientist, Genetic counselor

RESEARCH AREAS

Liquid biopsy cancer detection, Rapid infectious disease diagnostics, Companion diagnostic development

● CURRENT TRENDS

Growing development of 'liquid biopsy' tests that detect cancer DNA fragments from a simple blood draw.

● REAL-LIFE EXAMPLES

- ▶ COVID-19 PCR testing
- ▶ Genetic testing to select targeted cancer therapies
- ▶ Prenatal genetic screening

● LIMITATIONS

- ▶ Testing equipment and reagents can be costly
- ▶ Requires skilled technicians and quality control
- ▶ Results interpretation can require genetic counseling

INDUSTRY APPLICATIONS

Diagnostics companies, Hospitals, Pharmaceuticals (companion diagnostics), Public health

FUTURE SCOPE

Expanding use of next-generation sequencing is making comprehensive genetic testing faster and more affordable.

● SUMMARY

Molecular diagnostics detects disease at the genetic level, enabling earlier, more precise diagnosis and personalized treatment selection.